

ELEC5305 Project Feedback Two

MFCCs versus Pretrained Audio Embeddings: Robustness of Environmental Sound Classification

1 Project Title

MFCCs versus Pretrained Audio Embeddings: Robustness of Environmental Sound Classification

2 Student Information

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3 Project Feedback One – change and research questions

Based on the comments from the project feedback one, we modify the main focus of this project.

The original project mainly focus on comparing the classification accuracy of the different audio features and machine learning method in the Urbansound8k dataset. However, only finding the method which has the highest benchmark accuracy, it can not fully demonstrate the stability of the system when the recording conditions have changed.

Therefore, the revised project focuses on the robustness of different audio representations when acoustic and recording conditions change.

This project mainly compare the following two audio representations:

- conventional Mel-Frequency Cepstral Coefficients (MFCCs);
- pretrained YAMNet audio embeddings.

These two representations are evaluated by simple SVM classifier. Using similar downstream classifier can reduce the influence of the classifier complexity, and makes the comparison mainly focus on the audio representations itself.

Main Research Question

How robust are conventional MFCC features and pretrained deep audio embeddings to realistic acoustic and recording-channel perturbations in environmental sound classification?

Secondary Research Question

Can changes in the audio representation predict when classification performance will fail?

Therefore, this project not only compare the classification performance under the clean condition, but also study that how noise, reverberation and channel effects will change audio representation and how these changes affect the classification performance.

4 Current Progress and Methodology

This project uses the UrbanSound8k dataset, which includes 8732 labeling audio segments from 10 environmental sound types. This dataset also provides 10 pre-defined official folds.

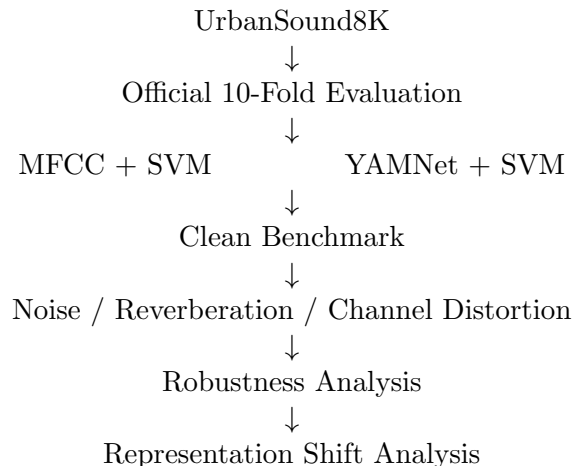
In order to avoid recreate random train-test split, this project will follow the official 10-fold evaluation protocol in all experiments.

For every fold:

1. use an official fold as held-out test set;
2. The left 9 folds are used for training;
3. repeat this process, until every folds all are used once test set.
4. Final results use the mean and standard deviation of 10 folds to report.

The whole process

Current project process is:



Reproducibility

A fixed random seed is used for the production of the results, to make the results reproducible. It also notes down the major experimental conditions such as MATLAB version used, preprocessing parameters, MFCC parameters, SVM parameters, and YAMNet loading parameters. These settings are automatically saved in a reproducibility manifest.

MFCC

MFCC baseline is design by a kind of interpretable signal-processing representation.

The main process phase of MFCC is:

Audio $\rightarrow$ Short-Time Frames $\rightarrow$ Windowing
 $\rightarrow$ Fourier Spectrum $\rightarrow$ Mel Filterbank
 $\rightarrow$ Log Energies $\rightarrow$ DCT $\rightarrow$ MFCC

The extracted MFCC features will combine with first-order deltas and use the statistical descriptive quantity to conclude, then input SVM to classify.

YAMNet

YAMNet is used as the pretrained feature extractor, rather than training from the scratch.

Its processing process:

Audio $\rightarrow$ Pretrained YAMNet $\rightarrow$ Embeddings
 $\rightarrow$ Statistical Summary $\rightarrow$ SVM

This method can conduct a controlled comparison between the traditional handcrafted audio representation with pretrained learned representation. The current implementation automatically checks and loads the pretrained YAMNet model before embedding extraction.

Current progress

current, I have complete the following contents:

- UrbanSound8K dataset preparation;
- metadata read and check the dataset;
- audio pre-processing;
- waveform, spectrum and spectrogram analysis;
- MFCC feature extraction;
- MFCC + SVM classification;
- pretrained YAMNet configuration;
- frozen YAMNet embedding extraction;
- YAMNet + SVM classification;
- official 10-fold evaluation;
- Accuracy and Macro-F1 calculation;
- per-class recall analysis;
- confusion matrix generate.

5 Preliminary Code and Results

The complete implementation code can be viewed in the GitHub repository. This section only presents a few representative parts of the current implementation.

Preliminary Code

MFCC feature extraction

The MFCC implementation first extracts frame-level MFCC coefficients and first-order deltas, then uses the statistical descriptive quantity to conclude these features.

MFCC firstly implement 13 frame-level MFCC coefficients and their first order delta. Then, these characteristics were statistically summarized using the mean, standard deviation, median, 25% quantile and 75% quantile. Finally, a fixed 130-dimensional representation is generated for each audio segment.

```
C = config();
x = preprocess_audio(x,fs,C.targetFs);

winLength = round(C.mfcc.frameMs/1000*C.targetFs);
hopLength = round(C.mfcc.hopMs/1000*C.targetFs);
overlapLength = winLength-hopLength;

M = mfcc(x,C.targetFs, ...
    "Window",hann(winLength,"periodic"), ...
    "OverlapLength",overlapLength, ...
    "NumCoeffs",C.mfcc.numCoeffs);

D = [zeros(1,size(M,2)); diff(M,1,1)];
F = [M D];

feat = [mean(F,1), std(F,0,1), median(F,1), ...
    prctile(F,25,1), prctile(F,75,1)];
```

YAMNet Embedding extraction

Pretrained YAMNet model keeps frozen status and only is used for feature extractor. By global average pooling layer, extracting 1024 dimensional embedding from each YAMNet patch. Then, using all the mean and standard deviation of patch embeddings to conduct statistical summary. Finally, a fixed 2048-dimensional representation is generated for each audio segment.

```
C = config();
x = preprocess_audio(x,fs,C.targetFs);

S = yamnetPreprocess(x,C.targetFs);

Z = predict(net,S, ...
    Outputs="global_average_pooling2d");

if isa(Z,"dlarray")
    Z = extractdata(Z);
end

Z = gather(Z);
Z = squeeze(Z);
```

```

if size(Z,1)==1024
    Z = Z.';
end

feat = [mean(Z,1), std(Z,0,1)];

```

Official 10-Fold Evaluation

The following part has achieved urbansound8k official evaluation procedure based on the folds.

```

for f = 1:10

    idxTest = folds==f;
    idxTrain = ~idxTest;

    Xtr = double(X(idxTrain,:));
    Xte = double(X(idxTest,:));
    Ytr = Y(idxTrain);
    Yte = Y(idxTest);

    % Training-only normalization
    mu = mean(Xtr,1);
    sigma = std(Xtr,0,1);
    sigma(sigma<1e-12) = 1;

    Xtr = (Xtr-mu)./sigma;
    Xte = (Xte-mu)./sigma;

    learner = templateSVM( ...
        "KernelFunction",C.svm.kernel, ...
        "BoxConstraint",C.svm.boxConstraint);

    model = fitcecoc(Xtr,Ytr, ...
        "Learners",learner,"Coding","onevsone");

    yPred = predict(model,Xte);
    M = evaluate_predictions(Yte,yPred,classOrder);
end

```

The initial results under the clean Condition

I have used official 10-fold evaluation protocol to complete clean-condition benchmark.

Table 1: The initial classification under the clean condition

method	avg Accuracy	avg Macro-F1
MFCC + SVM	63.43 $\pm$ 4.82%	0.642 $\pm$ 0.047
YAMNet + SVM	78.37 $\pm$ 3.19%	0.792 $\pm$ 0.037

Under the clean condition, pretrained YAMNet embeddings provide higher average classification performance than MFCC representation.

YAMNet improved the mean accuracy from 63.43 % to 78.37 %, and mean Macro-F1 increases from 0.642 to 0.792.

However, these results only build the clean-condition reference performance. They currently can not indicate that which representations are more robustness when the recording condition is changed.

Therefore, in the next stage, controlled perturbations applied to the held-out test recordings will be used to evaluate the two representations.

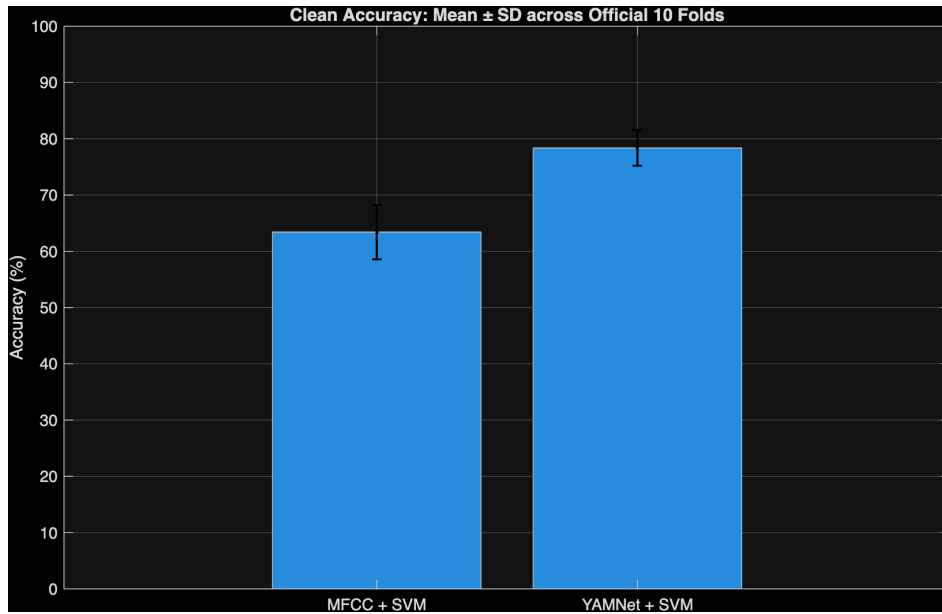


Figure 1: Average classification accuracy on official ten folds

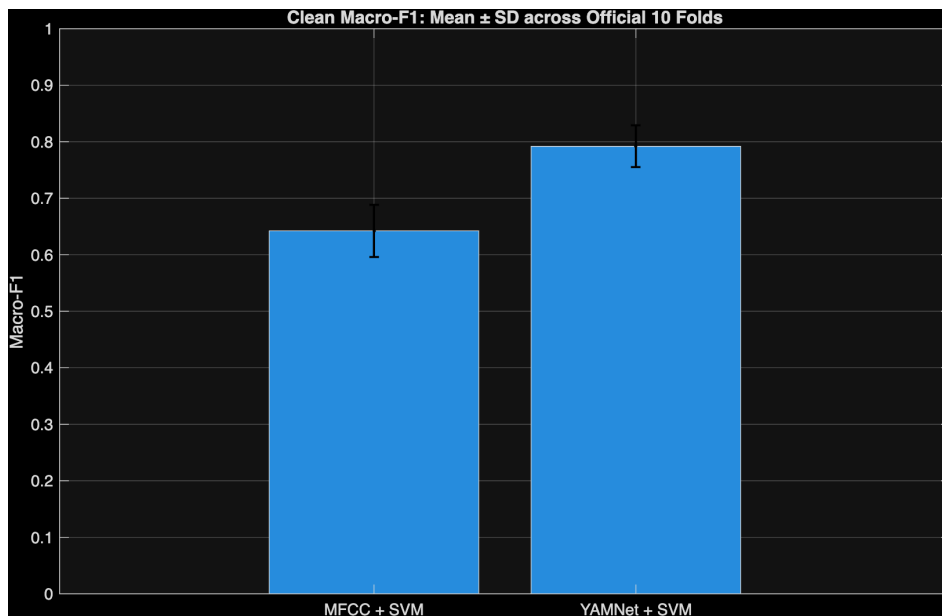


Figure 2: Average Macro-F1 on official ten folds

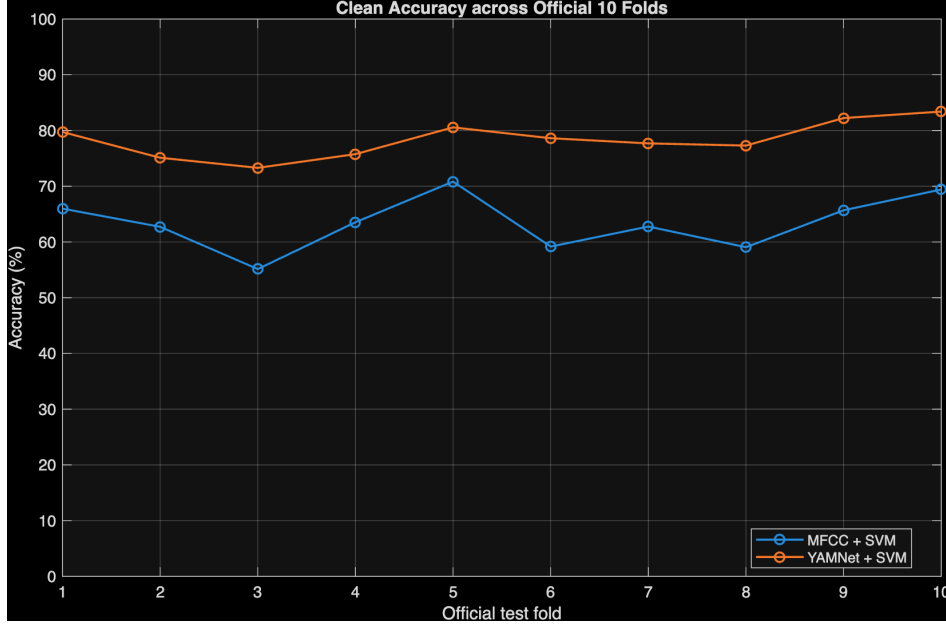


Figure 3: Ten Accuracy comparisons on official folds.

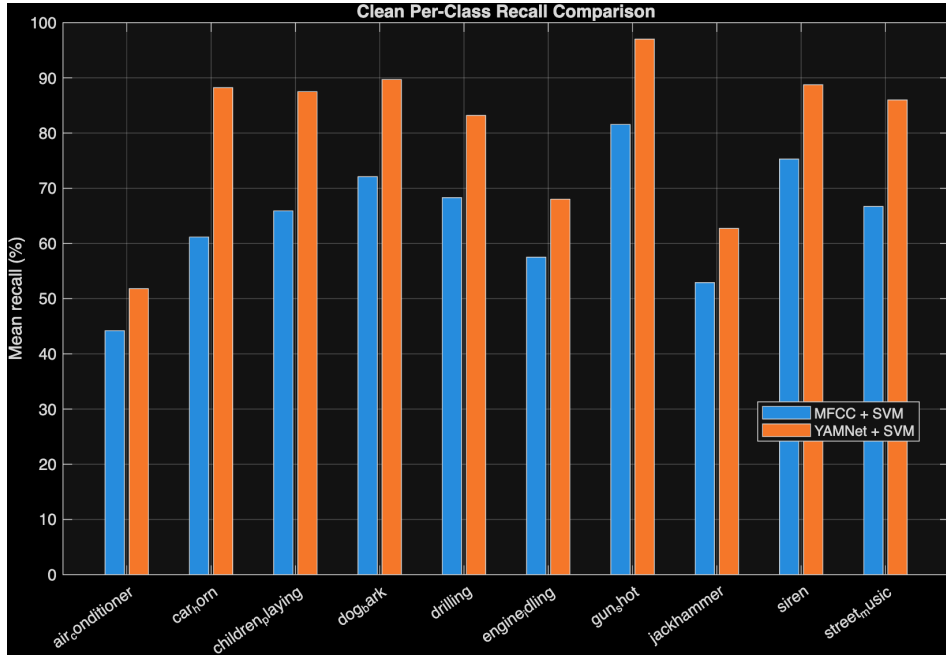


Figure 4: per-class recall comparison under Clean condition.

6 Planned Robustness and Representation Analysis

The next stage of this project is to study the performance of these two representations when the audio conditions or recording conditions are changed.

The train dataset will keep clean. Controlled perturbations only add to the held-out test recordings.

Both MFCC and YAMNet will use exactly same perturbed waveform to extract representations.

thereby ensuring that the primary experimental variable is the representation type.

Background Noise

The first robustness experiment will add the background noise under the controlled signal-to-noise ratios.

- 20 dB SNR;
- 10 dB SNR;
- 0 dB SNR

Accompanied by the noise level becoming more serious, the change of the classification performance will be evaluated.

The main result is to plot the Macro-F1 versus SNR of MFCC and YAMNet.

Reverberation

Add reverberation by convolving the test signal with the room impulse response:

$$y[n] = x[n] * h[n] \tag{1}$$

I will evaluate multiple reverberation levels, like mild, moderate and strong.

Channel and Frequency-Response Distortion

Using the controlled filtering to simulate the change of the recording-channel characteristics, like:

- microphone frequency-response differences;
- bandwidth limitation;
- recording or transmission channel variation.

Robustness Drop

For every representations, robustness drop is defined as:

$$\text{Robustness Drop} = \text{Clean Performance} - \text{Perturbed Performance} \tag{2}$$

This can compare the decline rate of two kinds of representation accompanying with the increase of perturbation severity.

Class-by-Class Analysis

The analysis will not only report the average performance.

For each sound class, we will compare the clean recall and perturbed recall:

$$\text{Clean Recall} \rightarrow \text{Perturbed Recall} \rightarrow \text{Recall Drop}$$

This can identify which sound classes are especially sensitive to noise, reverberation or channel distortion

Confusion matrices are also used to study how the error patterns change after perturbation.

If a specific confusion becomes more common, the relevant waveform, spectrum, spectrogram and MFCC representation will be further checked. Help interpret the results from the signal-processing perspective.

Representation Shift

This project will also measure how much the representation itself has changed after perturbation.

For MFCC features, the distance between the clean perturbed feature summaries will be calculated.

For YAMNet, it will use the cosine distance between the clean and perturbed embeddings.

Therefore, the main comparison is:

$$\text{Representation Shift} \longleftrightarrow \text{Classification Performance Drop}$$

Correlation Analysis

For each perturbation condition, compare the average representation shift with the corresponding Macro-F1 drop.

A simple correlation analysis will be used to investigate whether greater representation changes are associated with greater classification-performance degradation.

This analysis will directly answer the secondary research question.

7 Current Status and Next Steps

Table 2: Current project status.

Task	Status
Research question revision	Completed
UrbanSound8K preparation	Completed
Official 10-fold evaluation	Completed
MFCC + SVM	Completed
Frozen YAMNet + SVM	Completed
Clean Accuracy / Macro-F1	Completed
Per-class recall	Completed
Confusion matrices	Completed
Reproducibility setup	Completed
Background-noise robustness	Next (Week 10)
Reverberation robustness	Planned (Week 10)
Channel robustness	Planned (Week 11)
Class-by-class robustness	Planned (Week 11)
Representation-distance analysis	Planned (Week 12)
Correlation analysis	Planned (Week 12)

The current implementation also provides for an experiment setup that makes the experiment reproducible, fixed random seeds, automatic recording of experimental settings, and consistent clip-level identifiers.

At the current stage, the two clean-condition classification pipelines have been able to operate stably.

Next, the controlled background-noise experiment under multiple SNR levels will be implemented soon.

The reverberation and channel-distortion experiments will be added later.

At the final stage, the representation distance and class-level robustness will be analyzed. And the relationship between representation shift and classification-performance degradation will also be concluded.

8 References

References

- [1] J. Salamon, C. Jacoby, and J. P. Bello, “A dataset and taxonomy for urban sound research,” in *Proceedings of the 22nd ACM International Conference on Multimedia*, 2014.
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